

Core Pure Mathematics 2

Notes · Edexcel A Level Further Mathematics — Core Pure 2 · Chapter 7: Methods in Differential Equations

Chapter 7: Methods in Differential Equations

Why this matters

A Level Maths introduces separating variables for first-order differential equations. Further Maths builds on this in two ways: adding a second first-order technique (the **integrating factor**, for equations that can't be separated), and introducing **second-order linear differential equations with constant coefficients** — a completely new type of equation whose solution method (via an auxiliary equation) parallels solving quadratics, extended to the non-homogeneous case (a right-hand side $f(x) \neq 0$) using a particular integral. These methods together let you solve the great majority of differential equations that arise in mechanics and modelling.

Concept: separating variables (recap and extension)

For an equation of the form $dy/dx = f(x)g(y)$, we separate variables and integrate both sides:

$$\int 1/g(y) dy = \int f(x) dx$$

We assume familiarity with this from A Level Maths; here we simply extend it to more demanding functions f and g , and to finding particular solutions using given boundary/initial conditions.

Worked example 1 — Separating variables with a particular solution

WORKED EXAMPLE 1

Question. Solve the differential equation $(1+x^2) dy/dx = xy$, given that $y = 2$ when $x = 0$. Give y in terms of x .

Step 1. Separate variables: (M1) $\int 1/y dy = \int x/(1+x^2) dx$

Step 2. Integrate both sides (right-hand side by recognising the reverse chain rule, since the numerator is half the derivative of the denominator): (M1)(A1) $\ln|y| = \frac{1}{2} \ln(1+x^2) + c$

Step 3. Use the initial condition $y = 2$ at $x = 0$ to find c : (M1) $\ln 2 = \frac{1}{2} \ln 1 + c = 0 + c \implies c = \ln 2$

Step 4. Substitute back and solve for y : (M1) $\ln|y| = \frac{1}{2} \ln(1+x^2) + \ln 2 = \ln(1+x^2)^{1/2} + \ln 2 = \ln[2\sqrt{1+x^2}]$ So $y = 2\sqrt{1+x^2}$ (A1)

Answer: $y = 2\sqrt{1+x^2}$.

Concept: the integrating factor method

Some first-order equations cannot be separated because y and x are mixed additively rather than multiplicatively. If the equation can be written in the standard linear form:

$$dy/dx + P(x)y = Q(x)$$

then multiplying both sides by the **integrating factor** $I(x) = e^{\int P(x)dx}$ makes the left-hand side into the derivative of a product, allowing direct integration:

$$\frac{d}{dx} [I(x) \cdot y] = I(x) \cdot Q(x), \text{ so } I(x) \cdot y = \int I(x) Q(x) dx$$

Why this particular $I(x)$ works. This isn't a trick pulled from nowhere — it's designed backwards from what we want. By the product rule, $\frac{d}{dx}[I(x) \cdot y] = I(x) \cdot \frac{dy}{dx} + I'(x) \cdot y$. Compare this with $I(x) \cdot \frac{dy}{dx} + I(x)P(x)y$, which is what you get multiplying the original equation through by $I(x)$. These two expressions match exactly when $I'(x) = I(x)P(x)$ — itself a separable differential equation in I , which solves to $I(x) = e^{\int P(x)dx}$. So the integrating factor is simply *whatever function makes the product rule collapse the left-hand side back down to a single derivative* — that's the entire idea, and it's worth re-deriving this line if you ever forget the formula for $I(x)$.

Key formula

KEY FORMULA

Integrating factor: $I(x) = e^{\int P(x) dx}$, for $\frac{dy}{dx} + P(x)y = Q(x)$

Worked example 2 — Solving with an integrating factor

WORKED EXAMPLE 2

Question. Solve $\frac{dy}{dx} + (2/x)y = x^3$, given that $y = 1$ when $x = 1$.

Step 1. Identify $P(x) = 2/x$ and $Q(x) = x^3$ (equation is already in standard linear form). **(B1)**

Step 2. Find the integrating factor: **(M1)** $I(x) = e^{\int (2/x)dx} = e^{2 \ln x} = e^{\ln x^2} = x^2$

Step 3. Multiply the whole equation by $I(x) = x^2$: **(M1)** $x^2 \frac{dy}{dx} + 2xy = x^5$, i.e. $\frac{d}{dx}(x^2y) = x^5$

Step 4. Integrate both sides with respect to x : **(M1)** $x^2y = \int x^5 dx = x^6/6 + c$

Step 5. Apply the initial condition $y = 1$ at $x = 1$: **(M1)** $1 \cdot 1 = 1/6 + c \implies c = 5/6$

Step 6. Solve for y : **(A1)** $y = x^4/6 + 5/(6x^2)$

Answer: $y = x^4/6 + 5/(6x^2)$.

EXAM TIP

It is easy to forget to check the equation is in the standard form (coefficient of $\frac{dy}{dx}$ equal to 1) before reading off $P(x)$ — if the original equation is, say, $x \frac{dy}{dx} + 2y = x^4$, you must divide through by x first to get $\frac{dy}{dx} + (2/x)y = x^3$. Skipping this step is the single most common cause of a wrong integrating factor. Also note: after multiplying through by $I(x)$, the left-hand side *must* simplify to exactly $\frac{d}{dx}(I \cdot y)$ — if it doesn't, something has gone wrong earlier, so it's a great self-check.

Concept: second-order linear differential equations with constant coefficients (homogeneous case)

An equation of the form

$$a(\frac{d^2y}{dx^2}) + b(\frac{dy}{dx}) + cy = 0$$

(with a, b, c constants) is solved by trying a solution of the form $y = e^{mx}$, which leads to the **auxiliary equation**:

$$am^2 + bm + c = 0$$

The nature of the roots of this quadratic determines the form of the general solution:

Discriminant	Roots	General solution
$b^2 - 4ac > 0$	Two distinct real roots m_1, m_2	$y = Ae^{(m_1x)} + Be^{(m_2x)}$
$b^2 - 4ac = 0$	Repeated real root m	$y = (A + Bx)e^{(mx)}$
$b^2 - 4ac < 0$	Complex roots $p \pm qi$	$y = e^{(px)}(A \cos qx + B \sin qx)$

Worked example 3 — Second-order ODE with distinct real roots

WORKED EXAMPLE 3

Question. Find the general solution of $d^2y/dx^2 - 5(dy/dx) + 6y = 0$.

Step 1. Write down the auxiliary equation: **(M1)** $m^2 - 5m + 6 = 0$

Step 2. Solve by factorising: **(M1)** $(m - 2)(m - 3) = 0 \implies m = 2$ or $m = 3$

Step 3. Since the roots are real and distinct, write the general solution: **(A1)**

Answer: $y = Ae^{(2x)} + Be^{(3x)}$, where A and B are arbitrary constants.

Worked example 4 — Second-order ODE with a repeated root and a particular solution

WORKED EXAMPLE 4

Question. Find the particular solution of $d^2y/dx^2 + 4(dy/dx) + 4y = 0$, given that $y = 3$ and $dy/dx = 1$ when $x = 0$.

Step 1. Auxiliary equation: $m^2 + 4m + 4 = 0 \implies (m + 2)^2 = 0 \implies m = -2$ (repeated). **(M1)(A1)**

Step 2. General solution for a repeated root: **(A1)** $y = (A + Bx)e^{(-2x)}$

Step 3. Differentiate to apply the second boundary condition (product rule): **(M1)** $dy/dx = Be^{(-2x)} + (A + Bx)(-2)e^{(-2x)} = e^{(-2x)}[B - 2A - 2Bx]$

Step 4. Apply $y(0) = 3$: $A = 3$. **(M1)** Apply $y'(0) = 1$: $B - 2A = 1 \implies B - 6 = 1 \implies B = 7$

Answer: $y = (3 + 7x)e^{(-2x)}$. **(A1)**

EXAM TIP

The repeated-root case $(A + Bx)e^{(mx)}$ is the one most often got wrong by writing $y = Ae^{(mx)} + Be^{(mx)}$ (which is really just $(A+B)e^{(mx)}$ — a single arbitrary constant in disguise, since it can't satisfy two independent boundary conditions). Compare this with repeated roots of a quadratic to see why an extra x is needed: it's exactly analogous to needing a second, linearly independent solution, just as in a repeated-root quadratic factorisation you still only get one distinct value.

Worked example 5 — Second-order ODE with complex roots

WORKED EXAMPLE 5

Question. Find the general solution of $d^2y/dx^2 + 2(dy/dx) + 5y = 0$.

Step 1. Auxiliary equation: $m^2 + 2m + 5 = 0$. **(M1)**

Step 2. Use the quadratic formula: $m = [-2 \pm \sqrt{4 - 20}]/2 = [-2 \pm \sqrt{-16}]/2 = [-2 \pm 4i]/2 = -1 \pm 2i$. **(M1)(A1)**

Step 3. With complex roots $p \pm qi$ (here $p = -1$, $q = 2$), write the general solution: **(A1)**

Answer: $y = e^{-x}(A \cos 2x + B \sin 2x)$.

Concept: non-homogeneous second-order equations (particular integrals)

So far the right-hand side has always been 0 (a **homogeneous** equation). Edexcel Further Maths also expects you to solve the non-homogeneous case:

$$a(d^2y/dx^2) + b(dy/dx) + cy = f(x), \text{ where } f(x) \neq 0$$

The general solution has two parts: $y = (\text{complementary function}) + (\text{particular integral})$, often written **$y = CF + PI$** .

- The **complementary function (CF)** is the general solution of the associated homogeneous equation (set $f(x) = 0$) — exactly the method from the last few pages.
- The **particular integral (PI)** is *any single* function that satisfies the full equation with $f(x)$ on the right. You find it by trying a function of the same "family" as $f(x)$, with unknown coefficients, and substituting in to pin those coefficients down.

Why CF + PI works: if y_1 is any solution of the full equation and y_2 is any other solution, then $y_1 - y_2$ satisfies the *homogeneous* equation (substitute both in and subtract — the $f(x)$ cancels). So every solution of the full equation is (one particular solution) + (some solution of the homogeneous equation) — which is exactly $CF + PI$.

Key formula — standard trial functions for the particular integral

KEY FORMULA

$f(x)$ contains	Try PI of the form
a constant, or polynomial of degree n	a general polynomial of degree n (e.g. $f(x) = x^2 \rightarrow$ try $y = px^2 + qx + r$)
$ke^{\lambda x}$	$y = pe^{\lambda x}$
$k \cos(\omega x)$ or $k \sin(\omega x)$ (or a sum of both)	$y = p \cos(\omega x) + q \sin(\omega x)$ (always include <i>both</i> , even if $f(x)$ has only one)

Resonance case: if your trial PI (or any term of it) already appears in the complementary function, it will vanish when substituted in and you can't solve for its coefficient. Fix this by

multiplying the whole trial PI by x (or by x^2 if $x \cdot$ (that term) is *also* already in the CF).

Worked example 6 — Non-homogeneous second-order ODE

WORKED EXAMPLE 6

Question. Find the general solution of $d^2y/dx^2 - 5(dy/dx) + 6y = e^x$.

Step 1. Find the complementary function: the auxiliary equation $m^2 - 5m + 6 = 0$ factorises as $(m-2)(m-3) = 0$, so $m = 2$ or $m = 3$ (this is the same homogeneous equation as Worked example 3). **(M1)(A1)** CF: $y = Ae^{(2x)} + Be^{(3x)}$.

Step 2. Choose a trial PI: $f(x) = e^x$, and since e^x is not part of the CF (the CF only has $e^{(2x)}$ and $e^{(3x)}$), try $y = pe^x$, with $dy/dx = pe^x$ and $d^2y/dx^2 = pe^x$. **(M1)**

Step 3. Substitute into the full equation: $pe^x - 5pe^x + 6pe^x = e^x \rightarrow 2pe^x = e^x \rightarrow p = 1/2$. **(M1)(A1)**

Step 4. Combine CF and PI: **(A1)**

Answer: $y = Ae^{(2x)} + Be^{(3x)} + \frac{1}{2}e^x$.

EXAM TIP

A quick and valuable check: differentiate your PI twice and substitute it *alone* back into the original (non-homogeneous) equation — it should reproduce $f(x)$ exactly, independent of the CF. This isolates PI errors from CF errors, which is hard to do once they're added together. Also remember that arbitrary constants A and B are only ever attached to the CF, never to the PI — a PI is one specific function, not a family.

Resonance in action. Suppose instead the right-hand side had been $e^{(2x)}$ — the same exponent as one of the CF terms above. Trying $y = pe^{(2x)}$ would fail: substituting gives $p(4 - 10 + 6)e^{(2x)} = 0 \cdot pe^{(2x)} = 0$, which can never equal $e^{(2x)}$ whatever p is (the trial function solves the *homogeneous* equation, so it always substitutes to zero). The fix is to try $y = pxe^{(2x)}$ instead; differentiating this (product rule, twice) and substituting does produce a genuine $e^{(2x)}$ term you can match, giving a valid value of p . Recognising *why* the plain trial fails — it's secretly already a solution of the homogeneous equation — is more useful than memorising "multiply by x " as a rule.

Chapter summary

Idea	Key point
Separating variables	$\int 1/g(y) dy = \int f(x) dx$, for $dy/dx = f(x)g(y)$
Integrating factor	$I(x) = e^{\int P(x)dx}$ turns $dy/dx + Py = Q$ into $d/dx(Iy) = IQ$
Auxiliary equation	$am^2 + bm + c = 0$, from trying $y = e^{(mx)}$ in $ay'' + by' + cy = 0$
Real distinct roots	$y = Ae^{(m_1x)} + Be^{(m_2x)}$

Repeated root	$y = (A + Bx)e^{(mx)}$
Complex roots $p \pm qi$	$y = e^{(px)}(A \cos qx + B \sin qx)$
Non-homogeneous equations	$y = CF + PI$; try a PI matching the family of $f(x)$, multiply by x (or x^2) if it clashes with the CF

Common exam mistakes

- Not rearranging into standard form $dy/dx + Py = Q$ before reading off $P(x)$ for the integrating factor.
- Forgetting the constant of integration when integrating to find $I(x)y = \int IQ \, dx$ (though the $\int P \, dx$ step itself should NOT include a constant — only one $+c$ is needed overall, in the final integration).
- Writing $y = Ae^{(mx)} + Be^{(mx)}$ for a repeated root instead of the correct $(A + Bx)e^{(mx)}$.
- Sign errors solving the auxiliary equation with the quadratic formula, especially when the discriminant is negative.
- Forgetting to differentiate the general solution correctly (product rule) when applying a second boundary condition on dy/dx .
- Forgetting the particular integral entirely (giving only the CF as "the general solution"), or attaching arbitrary constants to the PI instead of only the CF.
- Choosing a trial PI that already appears in the CF without multiplying by x (or x^2) to fix the resonance clash.

You should now be able to

- Solve first-order differential equations by separating variables, including finding a particular solution from given conditions.
- Solve first-order linear differential equations using an integrating factor.
- Form and solve the auxiliary equation for a second-order linear homogeneous differential equation with constant coefficients.
- Write the correct general solution for real distinct roots, a repeated root, and complex roots, and find a particular solution using boundary conditions.
- Find the particular integral of a non-homogeneous second-order linear equation using the correct trial function, including the resonance case, and combine it with the complementary function.